

PHYS 143 – Mechanics and Thermal Physics
Week 1: Vectors – Tutorial Sheet

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Use the Master Procedure: identify, sketch, choose axes, predict, resolve, combine, reconstruct, evaluate. Give directions as angles measured counter-clockwise from $+x$ unless stated. Final answers to the practice problems and to the exam-style items are at the end.

A. Practice problems

A1. Magnitude and direction

Find the magnitude and the direction, measured counter-clockwise from $+x$, of $\vec{A} = (-6.0\hat{i} + 8.0\hat{j})$ m. Sketch first and name the quadrant.

A2. Two displacements

A walker goes 3.0 km east and then 4.0 km at 30° north of west. Find the resultant displacement (magnitude and direction from $+x$) using components.

A3. Dot and cross products

For $\vec{A} = (2.0\hat{i} + 3.0\hat{j} - 1.0\hat{k})$ and $\vec{B} = (1.0\hat{i} - 2.0\hat{j} + 4.0\hat{k})$ find $\vec{A} \cdot \vec{B}$, $\vec{A} \times \vec{B}$ and the angle between the vectors. Predict first whether the angle is acute or obtuse.

A4. Torque

A force of 60 N acts at the end of a 0.35 m spanner. The angle between the spanner and the force is 40° . Find the magnitude of the torque about the pivot.

A5. Work

A 50 N force at 25° above the horizontal drags a sledge 12 m along level ground. Find the work done by the force.

Exam-style practice

Ten five-option items in the style of the course examination, **suggested time: 3 minutes per item**. For each item sketch first, predict the sign or size, then calculate (IDENTIFY, SET UP, EXECUTE, EVALUATE). Choose one answer, A to E. Items marked as sharing information are linked. Answers and brief solutions are at the end of this sheet. Take $g = 9.80 \text{ m/s}^2$.

E1. A displacement $\vec{D}$ of magnitude 20.0 m points 30° south of west. Its x and y components ($+x$ east, $+y$ north), in metres, are

- (A) $(-17.3, -10.0)$
- (B) $(-17.3, 10.0)$
- (C) $(-10.0, -17.3)$
- (D) $(10.0, 17.3)$
- (E) $(17.3, -10.0)$

E2. $\vec{A} = (3.0\hat{i} - 5.0\hat{j})$ m and $\vec{B} = (-7.0\hat{i} + 2.0\hat{j})$ m. The direction of $\vec{A} + \vec{B}$, measured counter-clockwise from $+x$, is

- (A) 36.9° (B) 126.9° (C) 143.1° (D) 216.9° (E) 323.1°

E3. Which of the following statements is/are true for two non-zero vectors $\vec{A}$ and $\vec{B}$?

- I. $|\vec{A} + \vec{B}| = |\vec{A}| + |\vec{B}|$ only if $\vec{A}$ and $\vec{B}$ are parallel.
- II. $\vec{A} \times \vec{B} = \vec{B} \times \vec{A}$.
- III. $\vec{A} \cdot \vec{B}$ is a scalar and can be negative.

- (A) I only (B) I and II only (C) I and III only (D) II and III only (E) I, II and III

E4. For $\vec{P} = (\hat{i} - 3\hat{j} + 2\hat{k})$ and $\vec{Q} = (2\hat{i} + \hat{j} - \hat{k})$, the magnitude of $\vec{P} \times \vec{Q}$ is

- (A) 3.00 (B) 7.35 (C) 8.66 (D) 9.17 (E) 13.0

Items E5 and E6 share this information: A force $\vec{F} = (4.0\hat{i} - 2.0\hat{j} + 3.0\hat{k})$ N acts on a body that undergoes the displacement $\vec{d} = (2.0\hat{i} + 5.0\hat{j} - 1.0\hat{k})$ m.

E5. The work done by $\vec{F}$ during the displacement is

- (A) -18.0 J (B) -5.0 J (C) 5.0 J (D) 18.0 J (E) 29.5 J

E6. Using your previous result, the angle between $\vec{F}$ and $\vec{d}$ is closest to

- (A) 45° (B) 80° (C) 90° (D) 100° (E) 170°

E7. Three forces act at one point: 50.0 N along $+x$, 30.0 N at 120° from $+x$, and 40.0 N along $-y$. The magnitude of the resultant is

- (A) 19.0 N (B) 37.7 N (C) 52.0 N (D) 70.7 N (E) 120.0 N

E8. A force of magnitude 15 N is directed along $\vec{A} = (2\hat{i} - \hat{j} + 2\hat{k})$. The force, in newtons, is

- (A) $2\hat{i} - 1\hat{j} + 2\hat{k}$
- (B) $5\hat{i} - 5\hat{j} + 5\hat{k}$
- (C) $10\hat{i} - 5\hat{j} + 10\hat{k}$
- (D) $10\hat{i} + 5\hat{j} + 10\hat{k}$
- (E) $30\hat{i} - 15\hat{j} + 30\hat{k}$

E9. A force of 25.0 N acts at the end of a 0.4 m spanner. The angle between the spanner (tail to tail with the force) and the force is 150° . The magnitude of the torque $\vec{r} \times \vec{F}$ about the pivot is

- (A) 2.5 N m
- (B) 5.0 N m
- (C) 8.7 N m
- (D) 10.0 N m
- (E) 20.0 N m

E10. Vectors $\vec{A}$ and $\vec{B}$ have magnitudes 5.0 m and 3.0 m. Which value of $|\vec{A} + \vec{B}|$ is impossible?

- (A) 2 m (B) 4 m (C) 6 m (D) 8 m (E) 10 m

Answers to the practice problems

- A1 $|\vec{A}| = 10.0$; $\theta = 126.9^\circ$ (quadrant II, not -53.1°)
 A2 $R_x = -0.46$, $R_y = 2.00$ (km), $R = 2.05$ km at 103° from $+x$ (quadrant II)
 A3 $\vec{A} \cdot \vec{B} = -8$; $\vec{A} \times \vec{B} = 10\hat{i} - 9\hat{j} - 7\hat{k}$, magnitude 15.17; $\phi = 117.8^\circ$
 A4 $\tau = rF \sin \phi = 13.5$ N m
 A5 $W = Fs \cos \phi = 544$ J

Exam-style practice: answer key and brief solutions

Key: E1 A E2 D E3 C E4 C E5 B E6 D E7 B E8 C E9 B E10 E

E1 (A). Sketch first: south of west lies in quadrant III, so both components are negative. The 30° is measured from the west axis, so $D_x = -D \cos 30^\circ = -17.3$ and $D_y = -D \sin 30^\circ = -10.0$. Check: $\sqrt{D_x^2 + D_y^2} = 20.0$.

E2 (D). $\vec{R} = (-4.0\hat{i} - 3.0\hat{j})$ m: both components are negative, so $\vec{R}$ is in quadrant III (sketch). Reference angle $\tan^{-1}(3.0/4.0) = 36.9^\circ$, so $\theta = 180^\circ + 36.9^\circ = 216.9^\circ$. The calculator value 36.9° alone is the quadrant I trap.

E3 (C). I is true (the magnitudes add only when the vectors point the same way). II is false: $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$, so the order matters. III is true: $\vec{A} \cdot \vec{B} = AB \cos \phi$ is negative for $\phi > 90^\circ$. Answer: I and III only.

E4 (C). $\vec{P} \times \vec{Q} = [(-3)(-1) - (2)(1)]\hat{i} - [(1)(-1) - (2)(2)]\hat{j} + [(1)(1) - (-3)(2)]\hat{k} = \hat{i} + 5\hat{j} + 7\hat{k}$, so the magnitude is $\sqrt{1 + 25 + 49} = 8.66$. Check: it cannot exceed $PQ = 9.17$. The value $|PQ| = 9.17$ would need $\vec{P} \perp \vec{Q}$, but here $\vec{P} \cdot \vec{Q} = -3 \neq 0$.

E5 (B). $\vec{F} \cdot \vec{d} = (4.0)(2.0) + (-2.0)(5.0) + (3.0)(-1.0) = 8.0 - 10.0 - 3.0 = -5.0$ J. The negative sign means that, on balance, $\vec{F}$ has a component against the displacement.

E6 (D). $\cos \phi = \frac{\vec{F} \cdot \vec{d}}{Fd} = \frac{-5.0}{(5.385)(5.477)} = -0.170$, so $\phi = 99.8^\circ$: slightly obtuse, as the negative dot product predicted.

E7 (B). $R_x = 50.0 + 30.0 \cos 120^\circ = 35.0$ N, $R_y = 30.0 \sin 120^\circ - 40.0 = -14.0$ N, so $R = \sqrt{35.0^2 + 14.0^2} = 37.7$ N, pointing right and down (quadrant IV). Adding the magnitudes (120 N) is the usual error.

E8 (C). $|\vec{A}| = \sqrt{4 + 1 + 4} = 3$, so $\hat{A} = \frac{1}{3}(2\hat{i} - \hat{j} + 2\hat{k})$ and $\vec{F} = 15\hat{A} = 10\hat{i} - 5\hat{j} + 10\hat{k}$. Check: $\sqrt{100 + 25 + 100} = 15$. The vector $\vec{A}$ itself has magnitude 3, not 15.

E9 (B). $\tau = rF \sin \phi = (0.40)(25.0) \sin 150^\circ = 5.0 \text{ N m}$. The 150° is the angle between the vectors, and $\sin 150^\circ = \sin 30^\circ$, so only half of rF counts. Using $\cos$ instead of $\sin$ gives 8.7 N m , and omitting the angle gives 10.0 N m .

E10 (E). $|\vec{A} + \vec{B}|$ lies between $|A - B| = 2.0 \text{ m}$ (antiparallel) and $A + B = 8.0 \text{ m}$ (parallel). So 10 m is impossible; the others are all attainable.